

Introduction to Aircraft Propulsion

Class 1



Why?

It has been estimated that as many as three million people may be airborne at any given moment. Where gas turbine engine technology is responsible as the prime mover for the vast proportion of this figure and therefore of critical importance.

What is Propulsion?

- The word is derived from two Latin words: pro meaning before or forwards and pellere meaning to drive. Propulsion means to push forward or drive an object forward. A propulsion system is a machine that produces thrust to push an object forward.

Aircraft Propulsion

- Aircraft propulsion refers to the methods devised to produce a thrust force for the propulsion of a vehicle in flight, based on the principle of the momentum change of a fluid by the propulsion system. Principally the fluid being the gas used by the engine itself or the fluid available in the surrounding environment.

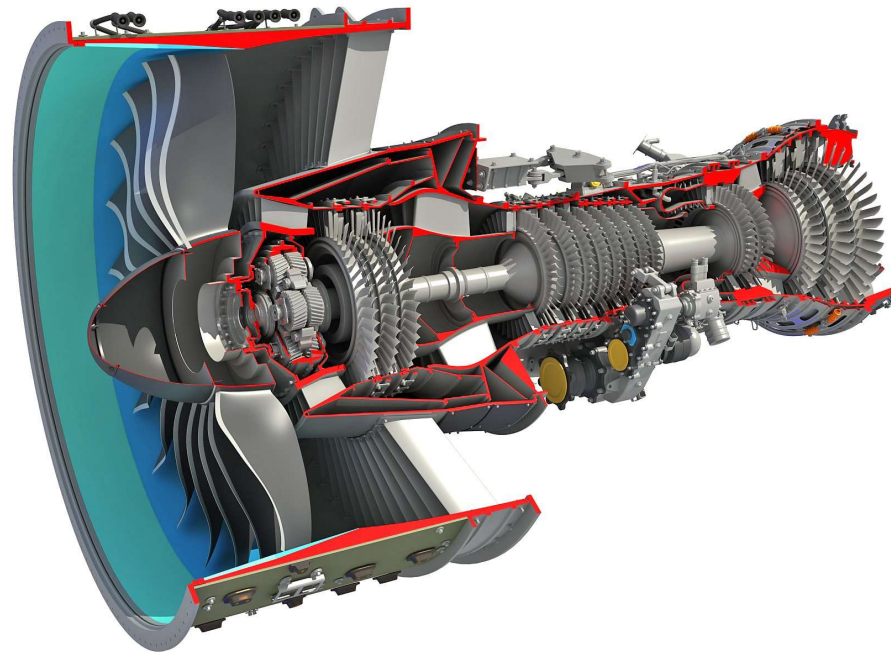
Aircraft propulsion

Airbreathing propulsion

Non – Airbreathing propulsion

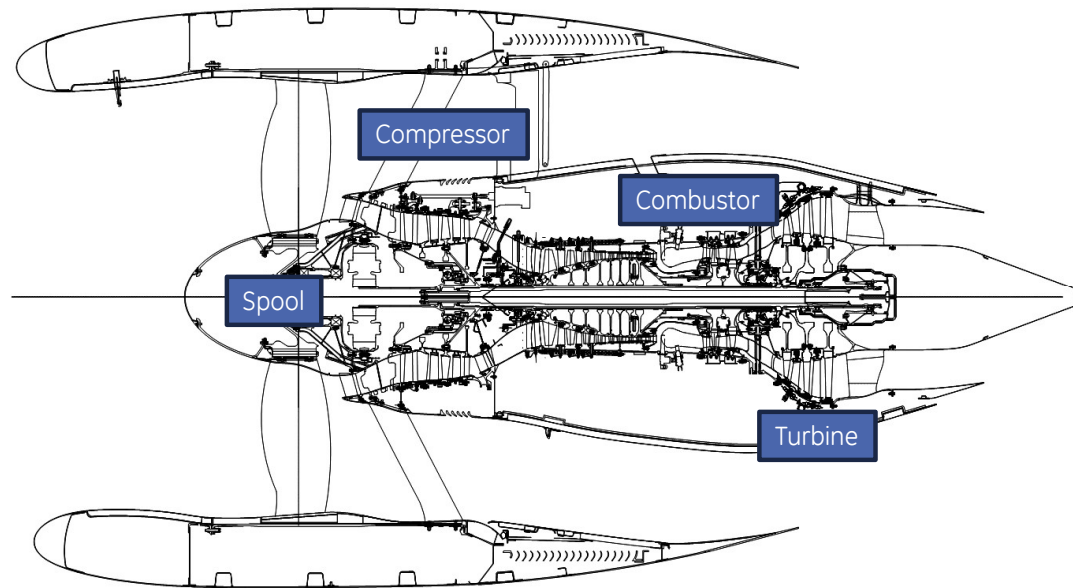
What is a gas turbine?

- A possible definition of a gas turbine is that: “A gas turbine is an entropic, nonadiabatic, unsteady, compressible flow, work extraction-producing machine”



Gas Turbine Main Parts

- Keywords:
 - Spool/shaft
 - Single/Multi/Twin
 - Compressor Blades
 - Compressor Vanes/Stator
 - Liners/Basket
 - Transitions
 - Turbine Blades/Buckets
 - Turbine Vanes/Nozzles
 - Exhaust
 - Inlet Guide vanes (IGVs)
 - Variable guide vanes/stators
 - Blade Cooling



Gas Turbine applications



Aircraft propulsion

As a mean of moving/propelling an aircraft.
Civil – Military – Helicopters



Marine

As a mean of moving/propelling a ship/boat



Industrial

As a mean of power generation (normally through the use of a generator)

- Heavy Duty
- Aero derivative

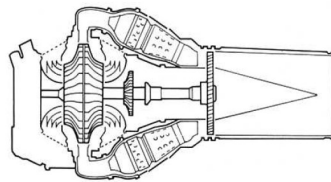


Auxiliary Power Units

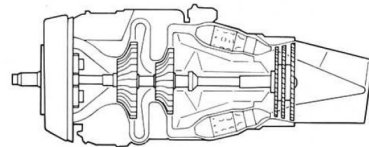
Micro/Small power generation technology

Gas Turbine Types

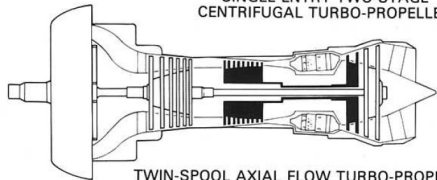
Turboprop



DOUBLE-ENTRY SINGLE-STAGE
CENTRIFUGAL TURBO-JET



SINGLE-ENTRY TWO-STAGE
CENTRIFUGAL TURBO-PROPELLER

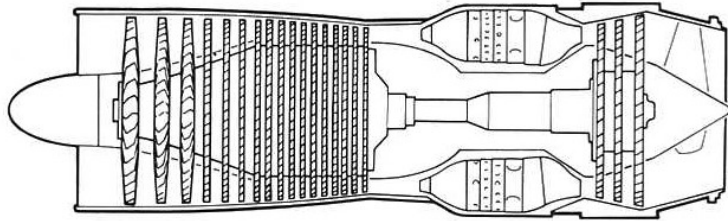


TWIN-SPOOL AXIAL FLOW TURBO-PROPELLER

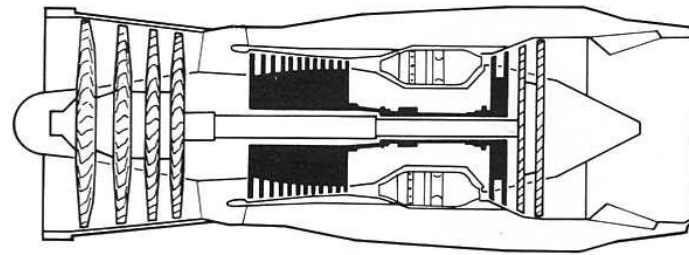


Gas Turbine Types

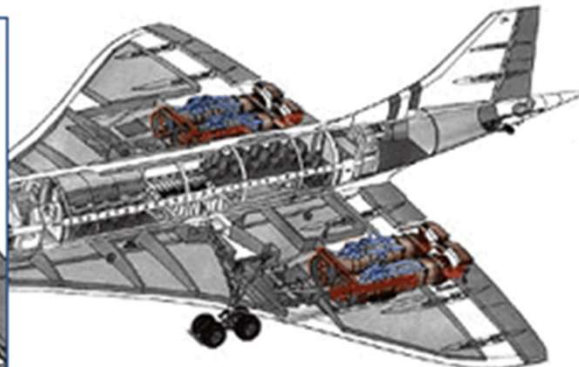
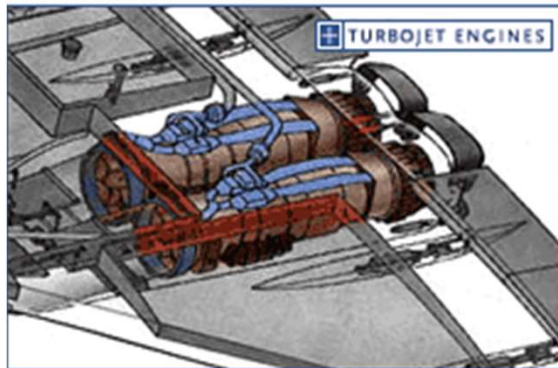
Turbojet



SINGLE-SPOOL AXIAL FLOW TURBO-JET

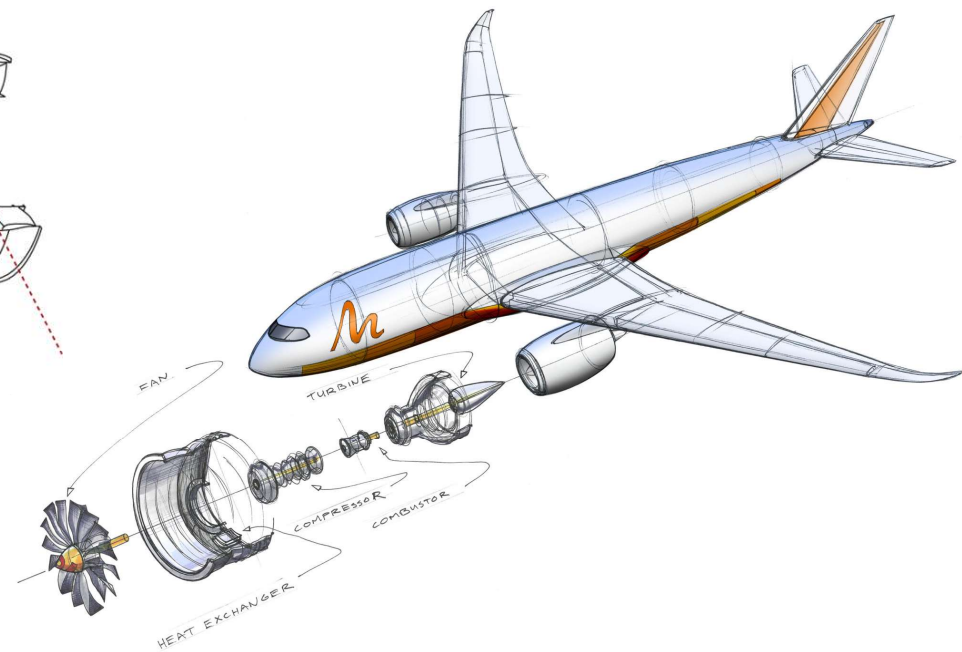
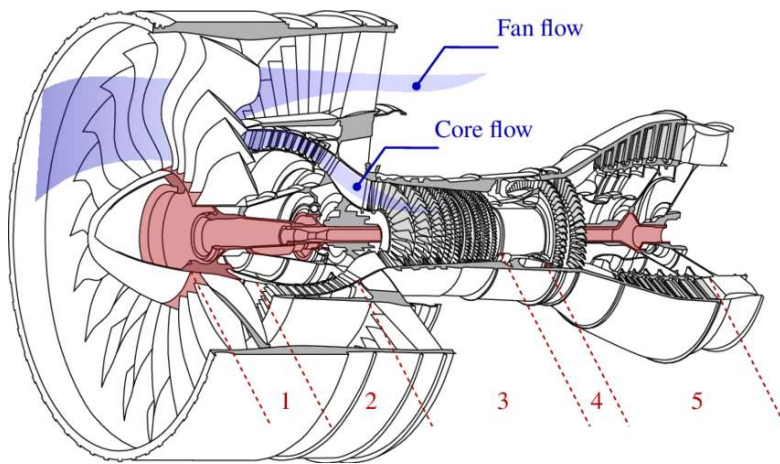


TWIN-SPOOL BY-PASS TURBO-JET
(low by-pass ratio)



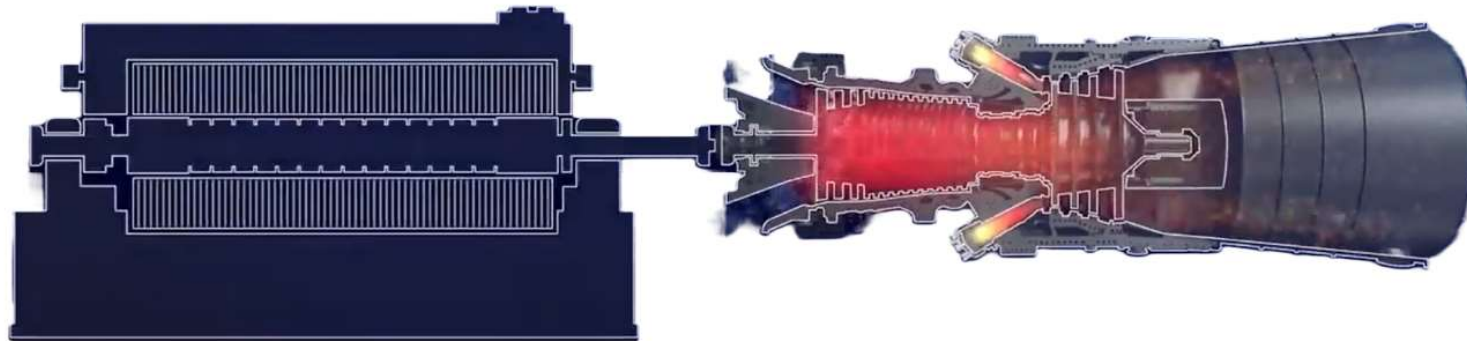
Gas Turbine Types

Turbofan



Gas Turbine Types

Heavy Duty – Power Generator

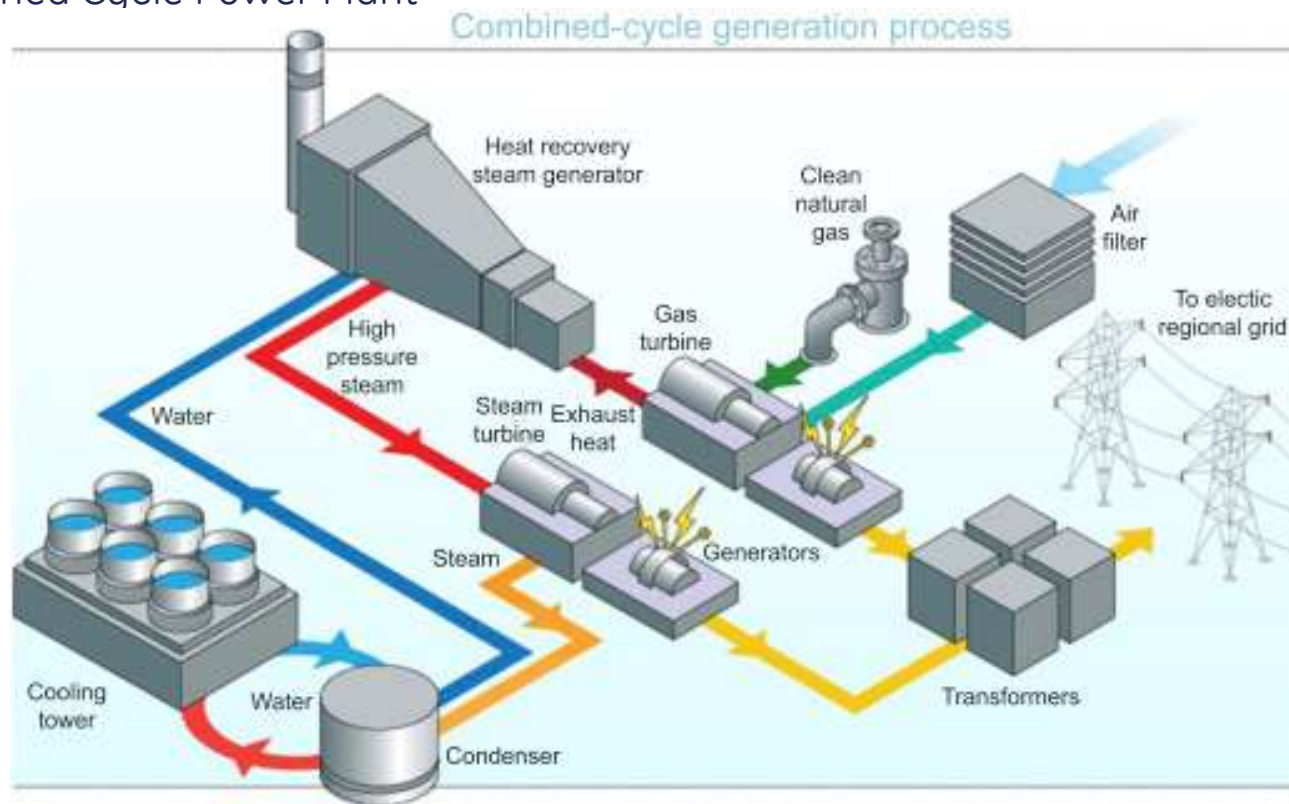


Generator

Gas Turbine

Gas Turbine Types

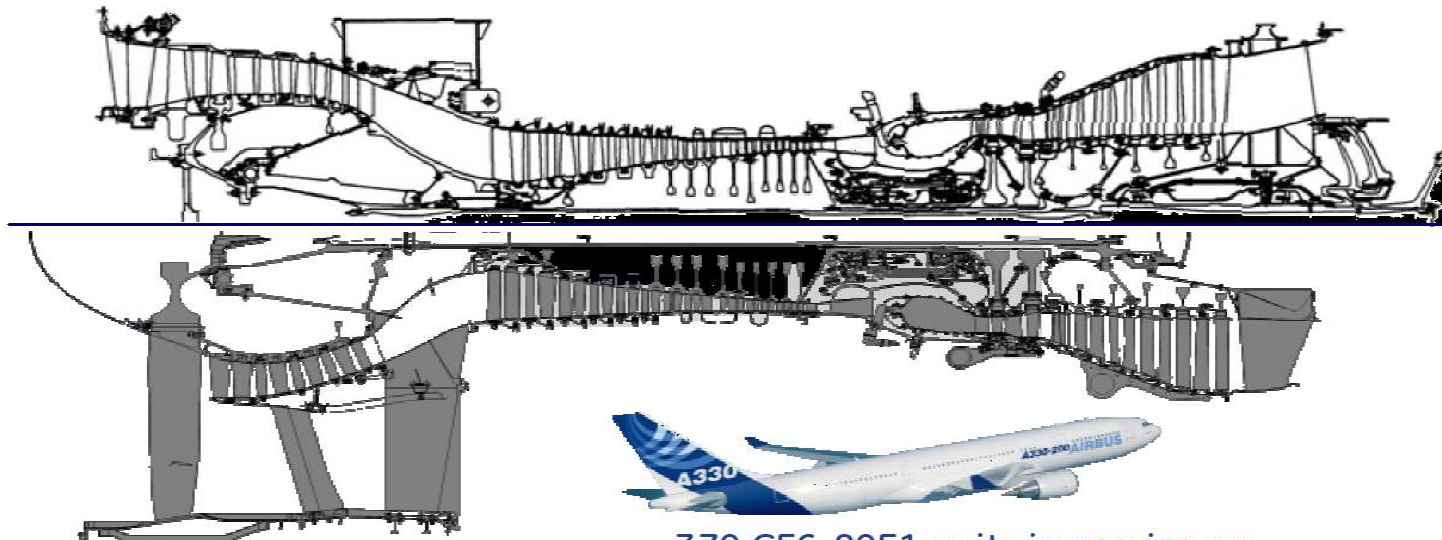
Combined Cycle Power Plant



Gas Turbine Types

Aeroderivative

LM6000PA

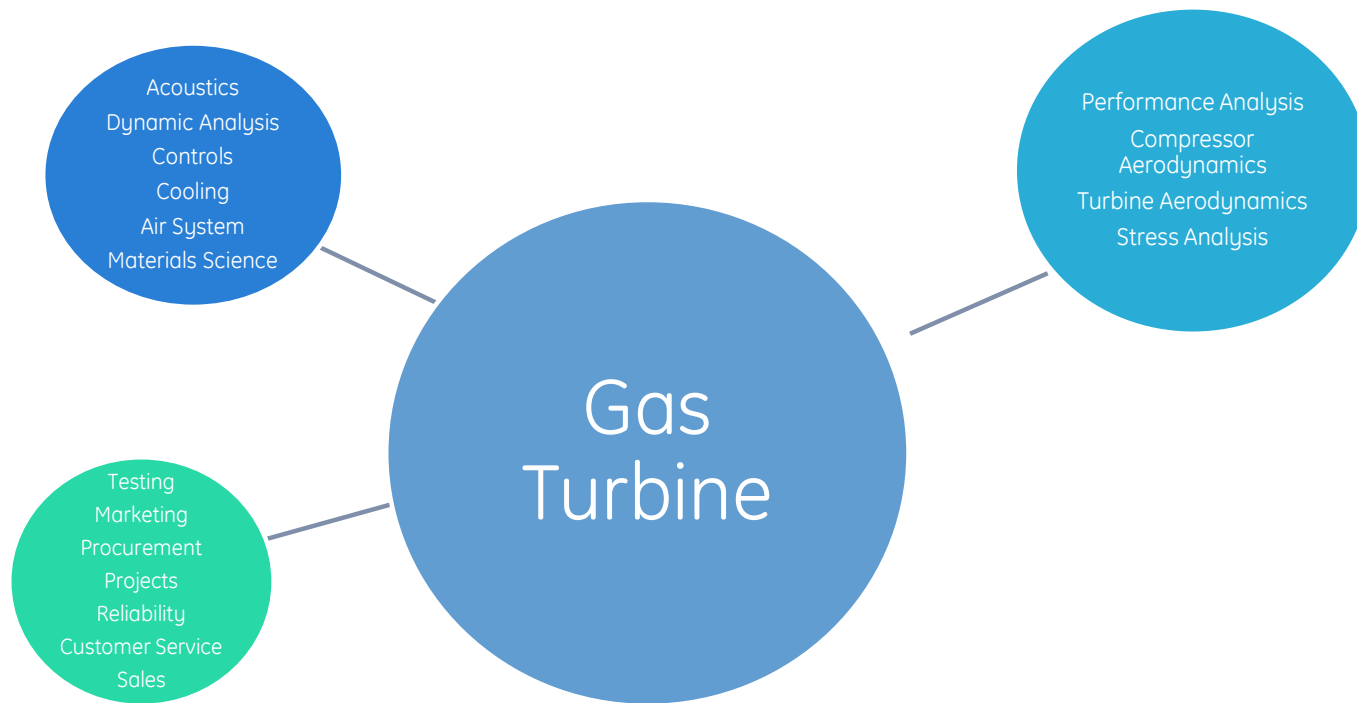


CF6-80



370 CF6-80E1 units in service on
Airbus A330 with 5M hours

Gas Turbine Technology Disciplines



Gas Turbine Fundamentals

Thermodynamics	Work (Euler equation)
----------------	-----------------------

	Enthalpy
--	----------

	Entropy
--	---------

	T-S diagrams
--	--------------

	Efficiency calculations
--	-------------------------

Fluid dynamics	Velocity Diagrams
----------------	-------------------

	Convergent – Divergent nozzles
--	--------------------------------

	Mach number
--	-------------

	Reynolds number
--	-----------------

Blade physics	Boundary layers
---------------	-----------------

	Wakes
--	-------

	Shocks
--	--------

	Vortices
--	----------

	Loss Models
--	-------------

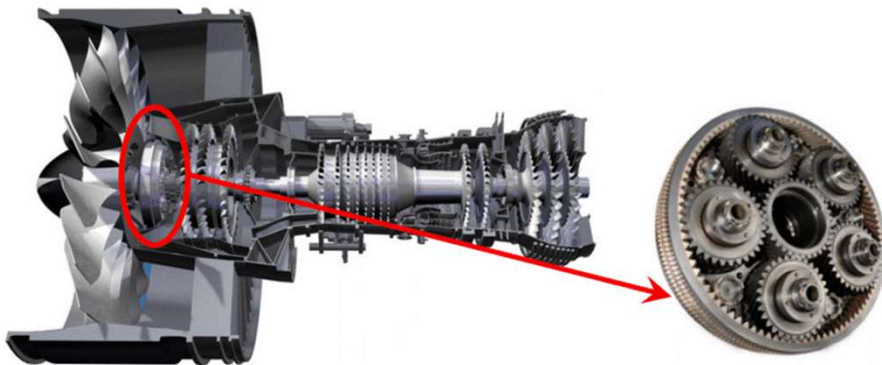
Gas Turbine Engineer

- What can I do?
 - Generally, a gas turbine engineer will participate in different activities such as model the gas turbine using different tools, methodologies and approaches, design the various components, optimize these components, respect aggressive schedules, interpret test results and data, and finally improve all this through various continuous improvement initiatives.
- Where can I work?

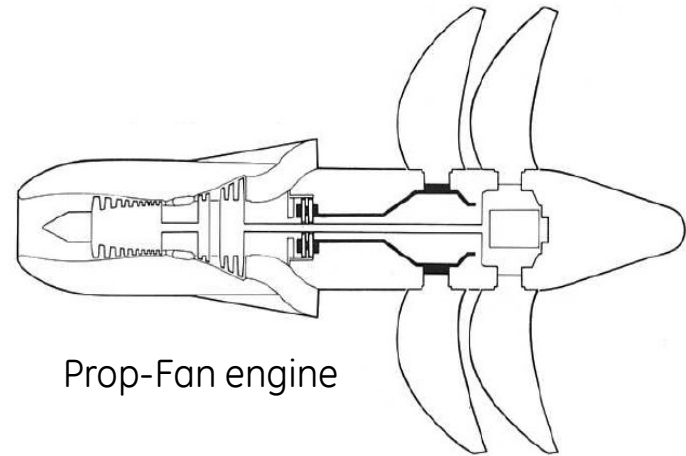


Gas Turbine Trends

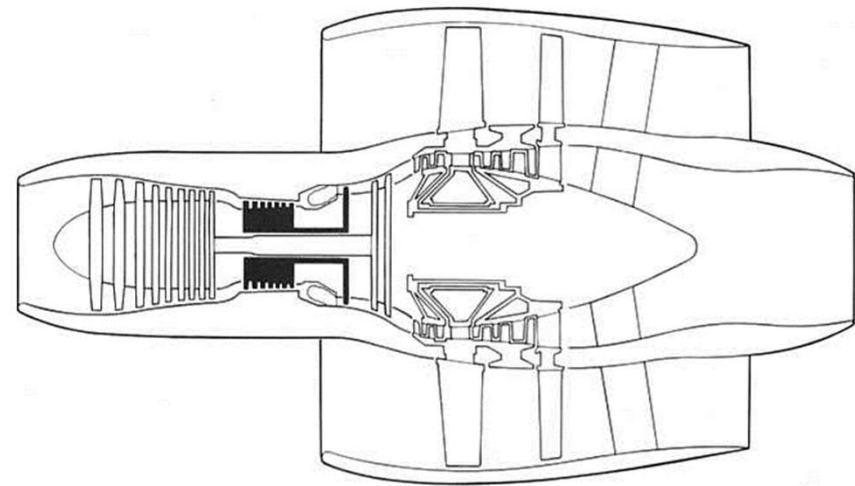
- Sustainable Aviation Fuel
- Additive manufacturing
- Higher firing temperatures
- Geared Turbofan/Advanced Turboprops



Geared turbofan engine

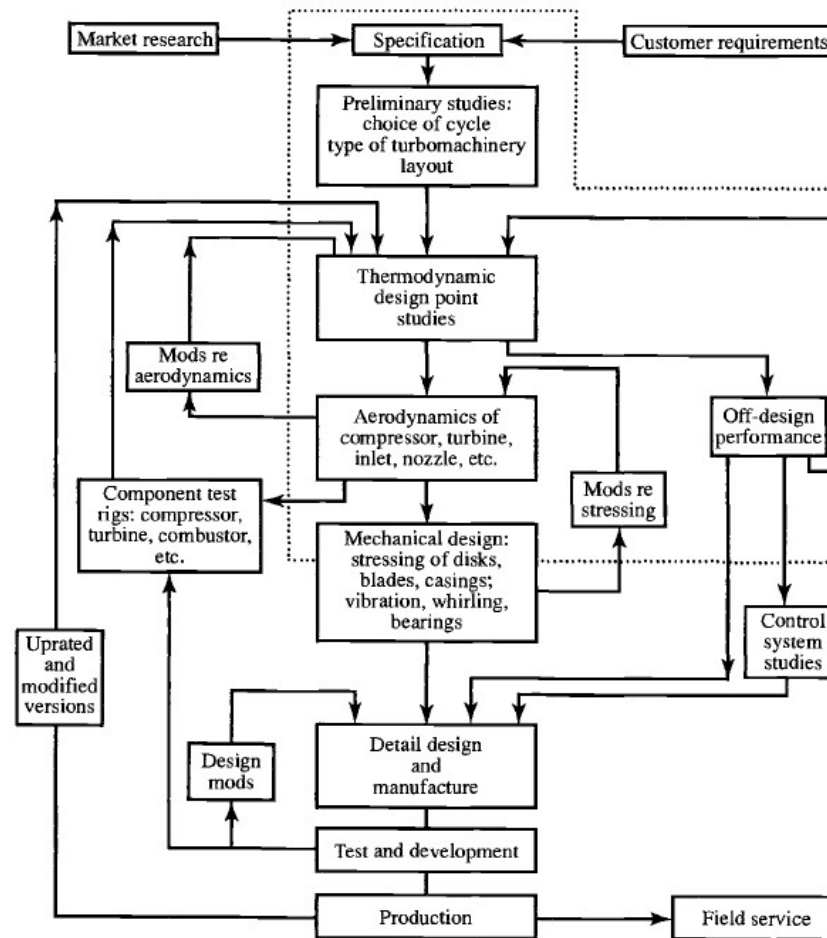


Prop-Fan engine



Contra-rotating fan engine

- There is no absolute roadmap for the design of a gas turbine engine. The design procedure for a typical gas turbine engine, shown in the picture, requires that thermodynamic design point studies and off design performance be the initial steps of design.



Reminder: Engineering Design and Analysis

- Design: Decision – making process englobing different fields such as engineering, economics and statistics applied iteratively to devise a system, system component, or process.
- Analysis: Part of a design process or problem-solving purpose, to form a coordinated whole.

